



Seasonal Beach-Profile Variation and Macroalgal Subsidies Shape Wrack-associated Macrofauna on Tropical Sandy Beaches of the Southwestern Atlantic

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Abstract

Sandy beaches are dynamic ecotones where beach-profile morphology, sediment properties, and allochthonous organic subsidies interact with intertidal benthic communities. We assessed seasonal variation in beach morphology, sediment granulometry, beach-cast macroalgal wrack, and associated macrofaunal assemblages at two tropical sandy beaches on the central Brazilian coast: Putiri Beach and Fifteen Beach. Both beaches were classified within the intermediate morphodynamic range according to Dean's dimensionless fall-velocity parameter, with $\Omega=3.29$ at Fifteen Beach and $\Omega=4.38$ at Putiri. Breaking wave conditions were comparable between beaches, with $H_b \approx 119.4$ cm and $T \approx 7.57$ s at Fifteen Beach and $H_b \approx 116.6$ cm and $T \approx 7.83$ s at Putiri. Sediment mean grain size remained within the fine-sand class at both beaches and in both seasons, but lower phi values in summer indicated relative coarsening within this class: Putiri shifted from phi 2.67 to phi 2.24, whereas 15 Beach shifted from phi 2.69 to phi 2.25. Sediment organic matter was also higher in summer than in winter, increasing from $2.81 \pm 0.68\%$ to $3.80 \pm 0.49\%$ at Putiri and from $2.87 \pm 0.39\%$ to $3.78 \pm 0.80\%$ at Fifteen Beach. Wrack biomass remained low and relatively stable at Putiri between winter and summer (53.81 ± 7.35 and 54.55 ± 1.05 g m⁻², respectively) and at Fifteen Beach in winter (56.93 ± 12.55 g m⁻²), but increased sharply at Fifteen Beach in summer (219.52 ± 83.31 g m⁻²), mainly due to brown algae and terrestrial plant material. Macrofaunal assemblages differed significantly among beach–season combinations, with Talitridae, Oligochaeta, and Cirolanidae together accounting for 98.4% of total abundance. Assemblage differences were supported by Bray–Curtis PERMANOVA, while relative-abundance patterns showed strong dominance of amphipods at Fifteen Beach in winter and increased contributions of oligochaetes and insects in summer assemblages. Together, these separate but complementary analyses indicate that beach–season combinations differed simultaneously in beach-profile morphology, sediment texture, wrack subsidies, and macrofaunal assemblage composition, supporting an association between physical habitat variation, trophic subsidies, and benthic assemblage structure.

Keywords Sandy beaches · Morphodynamics · Algal wrack · Macrofauna · Tropical Atlantic · Management · Central Brazilian coast, seasonality

Introduction

Sandy beaches dominate the world's ice-free shorelines, forming a critical ecotone between terrestrial and marine realms (Brown and McLachlan 2002; Schlacher et al. 2008). These dynamic ecosystems provide essential ecological services, including nutrient cycling, wildlife habitat, coastal protection, and recreation (Harris and Defeo 2022; Schlacher et al. 2007). Their physical environment is primarily governed by the interaction of waves, tides,

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and sediment supply, producing beach states that can be classified along the reflective–intermediate–dissipative continuum described by the Wright and Short morphodynamic model (Wright and Short 1984). This morphodynamic state acts as a master template, dictating swash climate, sediment stability, and ultimately, the availability and harshness of habitat for resident biota (McLachlan and Dörvlo 2005). Consequently, beach morphology and sediment characteristics (e.g., grain size, sorting) are recognized as first-order filters structuring macrofaunal communities, influencing species composition, abundance, biomass, and life-history traits (Barboza et al. 2018; Defeo and McLachlan 2005, 2013).

In contrast to the generally low in situ primary productivity of sand beaches, allochthonous organic subsidies, particularly beach-cast macrophyte wrack (algae and seagrasses), constitute a pivotal energy and nutrient input (Colombini and Chelazzi 2003; Dugan et al. 2003). This wrack, transported from adjacent marine ecosystems, accumulates at the drift line, creating biogeochemical hotspots that enhance microbial activity and provide critical food and shelter for a wide range of invertebrates, including talitrid amphipods, insects, and cirolanid isopods (Hyndes et al. 2022; Rodil et al. 2019). The decomposition and consumption of wrack can fuel detrital food webs, increase macrofaunal abundance, richness, and biomass, and provide prey resources for higher trophic levels such as shorebirds (Baring et al., 2018; Emery et al. 2025; Dugan et al. 2003; Hyndes et al. 2022). The quantity, quality (e.g., algal functional group), and persistence of this subsidy are not random, but are mediated by physical drivers. Wrack deposition, retention, and degradation are influenced by both external supply processes and local beach conditions, including wind direction, coastal orientation, wave exposure, swash conditions, beach morphology, wrack patch size, and the composition of stranded material (Kirkman and Kendrick 1997; Olabarria et al. 2007; Lastra et al. 2014). Thus, wrack availability on sandy beaches is best interpreted as the outcome of interactions between delivery from donor habitats and local retention and processing on the shore.

Although the general paradigms linking morphodynamics, wrack, and macrofauna have been extensively developed and validated in temperate microtidal beaches (Barboza and Defeo 2015; Defeo and McLachlan 2005), their applicability and relative importance in tropical systems remain less comprehensively explored. Tropical sandy beaches often experience lower seasonal thermal variability but may be subject to pronounced shifts in wave energy, rainfall, and terrestrial runoff, potentially altering sediment transport and subsidy delivery in

unique ways (Costa et al. 2019; McLachlan and Defeo 2018). Moreover, tropical coastlines such as the southwestern Atlantic frequently represent zones of high biogeographic overlap and biodiversity, where diverse algal source communities (e.g., rhodolith beds, coral reefs) can supply taxonomically and functionally diverse wrack subsidies (Amado-Filho et al. 2012; Floeter et al. 2008). The central Brazilian coast, particularly the state of Espírito Santo, exemplifies this context. It is a recognized marine biodiversity hotspot and a biogeographic transition zone, harboring rich reef systems (Mazzuco et al. 2019, 2020), extensive rhodolith beds (Stelzer et al. 2021), and a highly diverse macroalgal flora, making it a prime region to study the coupling of physical and biological drivers on sandy shores (Anderson et al. 2022; Coppo et al. 2024a, b, 2025; Francini-Filho et al., 2013; Vassoler et al. 2025).

Despite this ecological significance, integrated studies that simultaneously quantify seasonal changes in beach morphology, sedimentology, wrack biomass/composition, and macrofaunal assemblages are scarce for this region. This gap is critical given mounting anthropogenic pressures, including coastal development, mechanical beach cleaning, and the emerging industrial interest in harvesting beach-cast algae for fertilizer production production, a practice that may severely compromise ecological infrastructure and ecosystem functioning if not properly managed (Harris and Defeo 2022; Hyndes et al. 2022; Martins et al. 2024a, b). Effective conservation and sustainable management of sandy beach ecosystems require a mechanistic understanding of how physical drivers and trophic subsidies interact to shape biodiversity patterns across spatial and temporal scales.

This study addresses these knowledge gaps by investigating two proximate yet contrasting sandy beaches (Putiri and 15 Beach) on the central Brazilian coast. We adopt an integrated, seasonal approach to evaluate whether macrofaunal community structure varies among beach–season combinations in association with seasonal beach-profile morphology, sediment granulometry, and the availability and composition of algal wrack subsidies. Specifically, our objectives are to: (i) quantify and compare seasonal (austral winter vs. summer) changes in cross-shore beach profiles and sediment grain-size distributions; (ii) assess seasonal and spatial variability in the biomass and functional group composition of stranded algal wrack; and (iii) evaluate how these physical and resource gradients explain patterns in macrofaunal assemblage composition, diversity, and dominance. By elucidating these coupled dynamics, our work aims to provide a robust scientific basis for the monitoring and management of tropical sandy beaches, particularly in the face of growing exploitation pressures on beach wrack resources.

Materials and Methods

Study Area

The study was conducted along the central portion of the Brazilian coast (southwestern Atlantic), in the state of Espírito Santo, a well-known biogeographic transition zone where tropical reef systems, extensive rhodolith beds, and sandy-beach ecosystems occur in close proximity (Fig. 1). This coastal-shelf mosaic encompasses the Abrolhos Bank and adjacent shelf domains and has been repeatedly identified as an ecotone of biodiversity, with high species richness and endemism in rhodolith-associated assemblages and reef-associated fishes (Anderson et al. 2022; Coppo et al. 2024a, b). In the global Marine Ecoregions of the World framework, the area lies within the Tropical Southwestern Atlantic realm and the Eastern Brazil marine ecoregion (Spalding et al. 2007). The same central Brazilian shelf also supports the endemic deep-water kelp *Laminaria abyssalis*, whose restricted distribution and vulnerability to human pressures highlight the conservation relevance of this transition region (Anderson et al. 2025).

This coastal sector is also recognized as a national centre of marine macroalgal diversity and a key area for tropical–subtropical overlap in seaweed assemblages. Recent syntheses and regional inventories highlight high species richness and the conservation relevance of Espírito Santo

and adjacent central-eastern Brazil for beach-cast seaweeds and coastal macroalgal flora (Martins et al. 2024a, b; Vas-soler et al. 2025). The same region is repeatedly identified as a priority area for reef biodiversity and marine conservation planning based on hotspot-mapping approaches for reef fishes and reef habitats (Floeter et al. 2008; Mazzuco & Bernardino et al., 2022; Vila-Nova et al., 2014).

Field sampling targeted two nearby sandy beaches ~1 km apart from each other [Putiri and Fifteen Beach (hereafter referred to as 15 Beach)] within the central Brazilian coast, selected to represent contrasting seasonal beach-profile dynamics, sedimentological characteristics, and wrack-deposition patterns, while both beaches were classified within the intermediate morphodynamic range based on Dean's parameter (Coutinho and Bernardino 2017). Seasonal surveys were conducted in austral winters and summers of 2011 and 2012. Beach cross-shore morphology was quantified using the Emery profiling method (Emery 1961), which was used to describe seasonal beach-profile variation at each site. Morphodynamic state was classified separately using Dean's dimensionless fall-velocity parameter (Ω), calculated from wave height, wave period, and sediment fall velocity, following the Wright and Short (1984) morphodynamic framework. Beach morphology was characterized using the Emery profiling method (Emery 1961), and beach state was interpreted using the Wright and Short morphodynamic model, which classifies sandy beaches

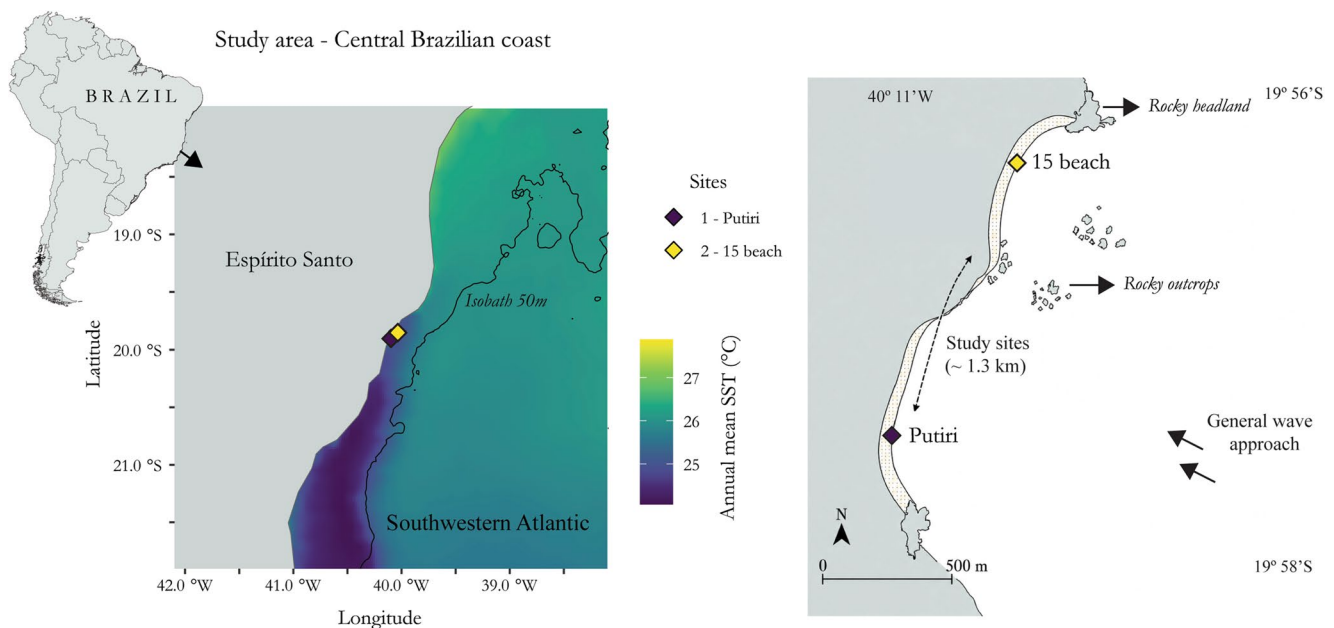


Fig. 1 Study area on the central Brazilian coast, Espírito Santo, Southwestern Atlantic. The left panel shows the regional geographic context of the two sampling sites, Putiri Beach and 15 Beach, including their position along the Espírito Santo coast, the spatial pattern of annual mean sea-surface temperature (SST, °C), and the 50-m isobath for

bathymetric reference. The right panel provides a local-scale representation of the sampled coastal sector, showing the relative position of Putiri Beach and 15 Beach, rocky headlands, rocky outcrops, the approximate distance between sites (~1.3 km), and the general wave approach

along the reflective–intermediate–dissipative continuum based on the coupling among wave conditions, sediment properties, and beach-profile morphology (Wright and Short 1984). Biomass was then classified into broad functional/taxonomic groups to characterize the composition and likely source categories of allochthonous inputs to the sandy-beach food web (Colombini and Chelazzi 2003; Liebowitz et al. 2016). In the laboratory, macrofauna associated with the wrack was separated under a stereomicroscope and identified to the smallest feasible taxonomic level, typically family. Because the sampling design targeted macrofauna associated with beach-cast wrack deposits, the dataset may include both resident sandy-beach fauna using wrack as habitat or food and organisms transported with stranded macrophytes from donor habitats; these origins were not separated experimentally. Sediment granulometry was assessed by separating samples into standard Wentworth size classes (gravel to mud) and weighing fractions to describe compositional shifts among beaches and seasons.

Sampling Design and Physical Characterization

Seasonal field surveys were conducted at Putiri Beach and 15 Beach during austral winter and summer campaigns in 2011 and 2012. At each beach and sampling campaign, three cross-shore sampling stations/blocks (B1–B3) were established along the sampled intertidal beach sector, from the upper beach/drift-line region toward the lower intertidal zone. Within each station/block, three replicate 50×50 cm quadrats (R1–R3; 0.25 m^2 each) were sampled for beach-cast wrack and associated macrofauna, resulting in nine wrack/macrofauna samples per beach per campaign. Quadrats were not randomly distributed across the entire cross-shore transect and were not placed at fixed distances independent of wrack occurrence. Rather, within each station/block, quadrats were positioned on visible beach-cast wrack deposits along the upper beach/drift-line, corresponding to the high-tide wrack accumulation zone. This targeted placement was adopted because the study specifically aimed to quantify stranded wrack biomass and the macrofauna associated with these deposits. Sediment samples for granulometric and organic-matter characterization were collected within the same beach $\times$ season sampling structure and were used as replicate sediment samples for each beach $\times$ season combination. Thus, wrack, associated macrofauna, and sedimentological conditions were assessed across the same station/block framework. Stranded macrophyte material within each quadrat was collected manually and stored in labelled bags. In the laboratory, associated fauna was separated from the wrack, and the remaining stranded material was sorted into broad functional/taxonomic categories: green algae,

brown algae, red algae, terrestrial plant material, and other algal material. Each category was oven-dried until constant mass and weighed to estimate dry biomass. At each beach and sampling period, replicated cross-shore Emery profiles were surveyed along the sampled beach sector to characterize local beach morphology. Each profile extended from the upper beach/drift-line region toward the lower intertidal zone, following the same general cross-shore orientation used for sediment, wrack, and macrofaunal sampling. Relative beach-profile elevation was calculated separately for each profile and then used to describe seasonal variation in beach morphology within each beach.

Beach cross-shore morphology was quantified using the Emery profiling method (Emery 1961). At each cross-shore profile, successive measurements were taken from the upper beach/drift-line region toward the lower intertidal zone using two graduated rods positioned at consecutive points along the transect. Horizontal distance between successive points was recorded, and relative differences in beach-surface elevation were estimated following the Emery method. These stepwise elevation differences were accumulated along each transect to obtain relative beach-profile elevation, using the initial point of each profile as the local reference. The resulting profiles describe relative cross-shore beach-surface elevation in meters and were used to characterize seasonal beach-profile variation at each site. Beach slope was not estimated as a separate variable; instead, profile shape and relative beach-profile elevation were used as descriptive measures of local beach morphology. Relative beach-profile elevation was calculated from cumulative stepwise height differences measured along each cross-shore transect using the Emery profiling method. The initial point of each profile was used as a local reference point; therefore, elevation values represent relative beach-surface elevation along each transect, not absolute elevation tied to a fixed geodetic datum. Because no differential GPS or fixed base-station survey was conducted, beach profiles were used to describe seasonal differences in relative profile shape and beach-face morphology, rather than to quantify absolute sediment accretion, erosion, or sediment-volume change.

Morphodynamic state was classified using Dean's dimensionless fall-velocity parameter (Ω), following the Wright and Short (1984) morphodynamic framework. Dean's parameter was calculated as:

$$\Omega = H_b / (W_s \times T)$$

where H_b is breaking wave height, W_s is sediment fall velocity, and T is wave period. Sediment fall velocity (W_s) was estimated from the median sediment grain diameter (D_{50}) using the Soulsby (1997) settling-velocity formulation for

non-cohesive sediments. Median grain size expressed in phi units was first converted to grain diameter in millimetres using:

$$D50 = 2(-\phi)$$

The resulting D50 value was then converted to meters. The dimensionless grain-size parameter was calculated as:

$$Dstar = D50 \times [gx(s - 1)/\nu^2]^{(1/3)}$$

where Dstar is the dimensionless grain-size parameter, g is gravitational acceleration, s is the sediment-to-water density ratio, and ν is the kinematic viscosity of seawater. Sediment fall velocity was then estimated as:

$$Ws = (\nu/D50) \times \{[10.362 + 1.049 \times Dstar^3]^{(1/2)} - 10.36\}$$

Both beaches were then classified along the reflective–intermediate–dissipative continuum based on the resulting Omega values. Based on Ω , both beaches were classified within the intermediate morphodynamic range. Sediment samples were collected from the same beach $\times$ season sampling structure used for the biological surveys. Sediment granulometry was determined by separating dried sediment into standard Wentworth size classes, from gravel to mud, and weighing each fraction. Mean grain size was calculated in phi units from the proportional contribution of each sediment fraction. Total organic matter content was estimated by loss-on-ignition, using the percentage mass loss between dried sediment and combusted sediment:

$$TOM (\%) = \frac{[(dry\ sediment\ mass - ashfree\ sediment\ mass)]}{dry\ sediment\ mass} \times 100$$

Statistical Analysis

Beach-profile variation was analyzed using only the Emery profile dataset. Each data point corresponded to a relative beach-profile elevation measurement at a known cumulative distance along a cross-shore transect. Relative beach-profile elevation was modeled as the response variable using a Gaussian GLM, with beach, season, and their interaction as fixed effects, and centered cumulative distance as a continuous covariate to account for the systematic change in elevation along the beach profile (Nelder and Wedderburn 1972). The model was used to evaluate differences in profile elevation among beach $\times$ season combinations, whereas morphodynamic state was classified separately using Dean's parameter. Type-III tests, DHARMA diagnostics,

and estimated marginal means were used for inference and post-hoc comparisons (Fox and Weisberg 2019; Hartig 2022; Lenth 2025). Full outputs for the beach-profile GLM, including Type-III tests, model coefficients, estimated marginal means, post-hoc comparisons, compact-letter displays, and DHARMA diagnostics, are provided in Supplementary Table S1. 15Wrack biomass analyses were based on the replicated wrack samples collected within each beach $\times$ season combination. For each replicated sampling unit, stranded material was sorted into functional/taxonomic categories (green algae, brown algae, red algae, terrestrial plant material, and other algae) dried to constant mass, and expressed as biomass per unit area ($g\ m^{-2}$). Thus, each observation in the biomass dataset corresponded to the dry biomass of one wrack category within one replicated sampling unit. Zero values occurred when a given wrack category was absent from a particular sampling unit, but these zeros were retained because they represent true absences within the replicated design. To stabilize variance and allow inclusion of zero values, biomass was $\log_{10}$ -transformed after adding a small constant. Biomass was then analyzed using Gaussian generalized linear mixed models (GLMMs) fitted with glmmTMB (Brooks et al. 2017). Fixed effects included wrack group, beach, season, and their interactions, while sampling unit was included as a random intercept to account for non-independence among wrack categories measured within the same replicate. Type-III tests were used for inference on fixed effects (Fox and Weisberg 2019). Model assumptions were evaluated using simulation-based residual diagnostics in DHARMA (Hartig 2022). Estimated marginal means (EMMs) were computed for beach $\times$ season combinations within each wrack group, followed by Tukey-adjusted pairwise comparisons and compact letter displays (Hothorn et al. 2008; Lenth 2025). Full outputs for the wrack-biomass GLMM, including fixed-effect tests, model coefficients, estimated marginal means, Tukey-adjusted post-hoc comparisons, compact-letter displays, and DHARMA diagnostics, are provided in Supplementary Table S2. Granulometric analyses were based on replicated sediment samples collected within each beach $\times$ season combination. Each sediment sample was dried and separated into standard Wentworth grain-size classes, from mud to gravel, and the mass of each fraction was recorded (Wentworth, 1922). For ecological interpretation, sediment texture was summarized primarily using sample-level granulometric descriptors, including mean grain size, sorting, skewness, and the proportional contribution of major sediment fractions. Because sediment fractions are compositional by nature, the mass of one fraction within a sample is not fully independent from the others. Therefore, the fraction-level GLMM was used only as a descriptive inferential tool to summarize variation in sediment size-class structure among

beach $\times$ season combinations, rather than as a fully compositional model. Fraction masses were log₁₀-transformed to stabilize variance and analyzed using Gaussian generalized linear mixed models fitted with glmmTMB (Bolker et al. 2009; Brooks et al. 2017). Fixed effects included sediment size class, beach, season, and their interactions, and sample identity was included as a random intercept to account for repeated measurements of multiple sediment fractions within the same sediment sample. Model assumptions were evaluated using DHARMA simulation-based residual diagnostics (Hartig 2022), and estimated marginal means were compared using Tukey-adjusted post-hoc contrasts with compact letter displays (Hothorn et al. 2008; Lenth 2025). Accordingly, statistical results for individual grain-size fractions were interpreted cautiously and in combination with sample-level granulometric descriptors. Full outputs for the sediment-granulometry GLMM are provided in Supplementary Table S3. Sample-level and beach $\times$ season summaries of Folk and Ward granulometric descriptors, including mean grain size, sediment class, sorting, skewness, and the relative contribution of major sediment fractions, are provided in Supplementary Table S5. Species-by-station abundance matrices were compiled by summing organism counts within each sampled station, nested within beach and season. Community dissimilarity among stations was quantified using the Bray–Curtis index (Bray and Curtis 1957). Differences in assemblage composition were tested using a one-factor PERMANOVA implemented with adonis2 in vegan (Anderson 2001a, b; Oksanen et al. 2024), with beach $\times$ season combination as the fixed factor. This factor had four levels: Putiri–winter, Putiri–summer, 15 Beach–winter, and 15 Beach–summer. This simplified design was used to directly evaluate compositional differences among the sampled beach–season combinations while avoiding ambiguity associated with partitioning separate main effects and interaction terms. Pairwise PERMANOVA comparisons among beach $\times$ season groups were then performed, with p-values adjusted for multiple testing using the Benjamini–Hochberg false discovery rate (Benjamini and Hochberg 1995). Significance-grouping letters for pairwise tests were generated from adjusted p-values using multcompView (Piepho 2004). Because the objective of this study was to characterize macrofauna associated with wrack deposits and sampled beach–season stations, we did not conduct targeted coring or excavation of deeply burrowing lower-intertidal/swash-zone fauna. Therefore, taxa such as mole crabs (Hippoidea, e.g. *Emerita brasiliensis*), which typically occur buried in sandy sediments of the intertidal/swash zone, may be under-represented or absent from our samples. Full outputs for the faunal assemblage multivariate analyses, including global and pairwise PERMANOVA results are provided in Supplementary Table S4.

Percent relative abundances were converted to proportions and transformed to lie in (0, 1) using the Smithson–Verkuilen adjustment prior to fitting beta-distributed GLMMs (Smithson and Verkuilen 2006). Models included beach, season and their interaction as fixed effects, with random intercepts for taxonomic group and sample to account for repeated measures across taxa and sampling units. Residual diagnostics were assessed with DHARMA (Hartig 2022). Post-hoc EMM comparisons among beach $\times$ season combinations used Tukey adjustments and compact letter displays (Hothorn et al. 2008; Lenth 2025).

Complete statistical outputs for all models, including fixed-effect tests, model coefficients, estimated marginal means, post-hoc comparisons, compact-letter displays, and DHARMA residual diagnostics where applicable, are provided in Supplementary Tables S1–S4 (Davis and Keppel, 2021) while the granulometric descriptors are provided in Supplementary Table S5.

Results

Both beaches were classified within the intermediate morphodynamic range according to Dean’s dimensionless fall-velocity parameter, with $\Omega=3.29$ at 15 Beach and $\Omega=4.38$ at Putiri. Breaking wave conditions were comparable between beaches, with $H_b \approx 119.4$ cm and $T \approx 7.57$ s at 15 Beach and $H_b \approx 116.6$ cm and $T \approx 7.83$ s at Putiri. Sediment mean grain size remained within the fine-sand class at both beaches and in both seasons. However, lower phi values in summer indicated relative coarsening within the fine-sand range: Putiri shifted from ϕ 2.67 in winter to ϕ 2.24 in summer, whereas 15 Beach shifted from ϕ 2.69 to ϕ 2.25. Sediment organic matter content was also higher in summer than in winter, increasing from $2.81 \pm 0.68\%$ to $3.80 \pm 0.49\%$ at Putiri and from $2.87 \pm 0.39\%$ to $3.78 \pm 0.80\%$ at 15 Beach. Therefore, subsequent beach–season contrasts are interpreted as seasonal variation in beach-profile morphology, sediment texture, organic-matter content, and wrack subsidies, rather than as differences in categorical morphodynamic state. Relative beach-profile elevation also showed pronounced seasonal differences and clear beach $\times$ season contrasts (Fig. 2a). At Putiri, both seasons exhibited a consistently increasing cross-shore elevation, with winter showing a steeper and higher profile than summer. At 15 Beach, winter profiles were also steep and elevated, whereas summer profiles were markedly flattened with low relative beach-profile elevation across the transect, indicating substantial seasonal reworking of the beach-face.

These patterns were supported by the Gaussian GLM, which indicated a clear beach $\times$ season signal in predicted corrected height (Fig. 2b). Post-hoc comparisons of estimated marginal means showed that 15 Beach in summer

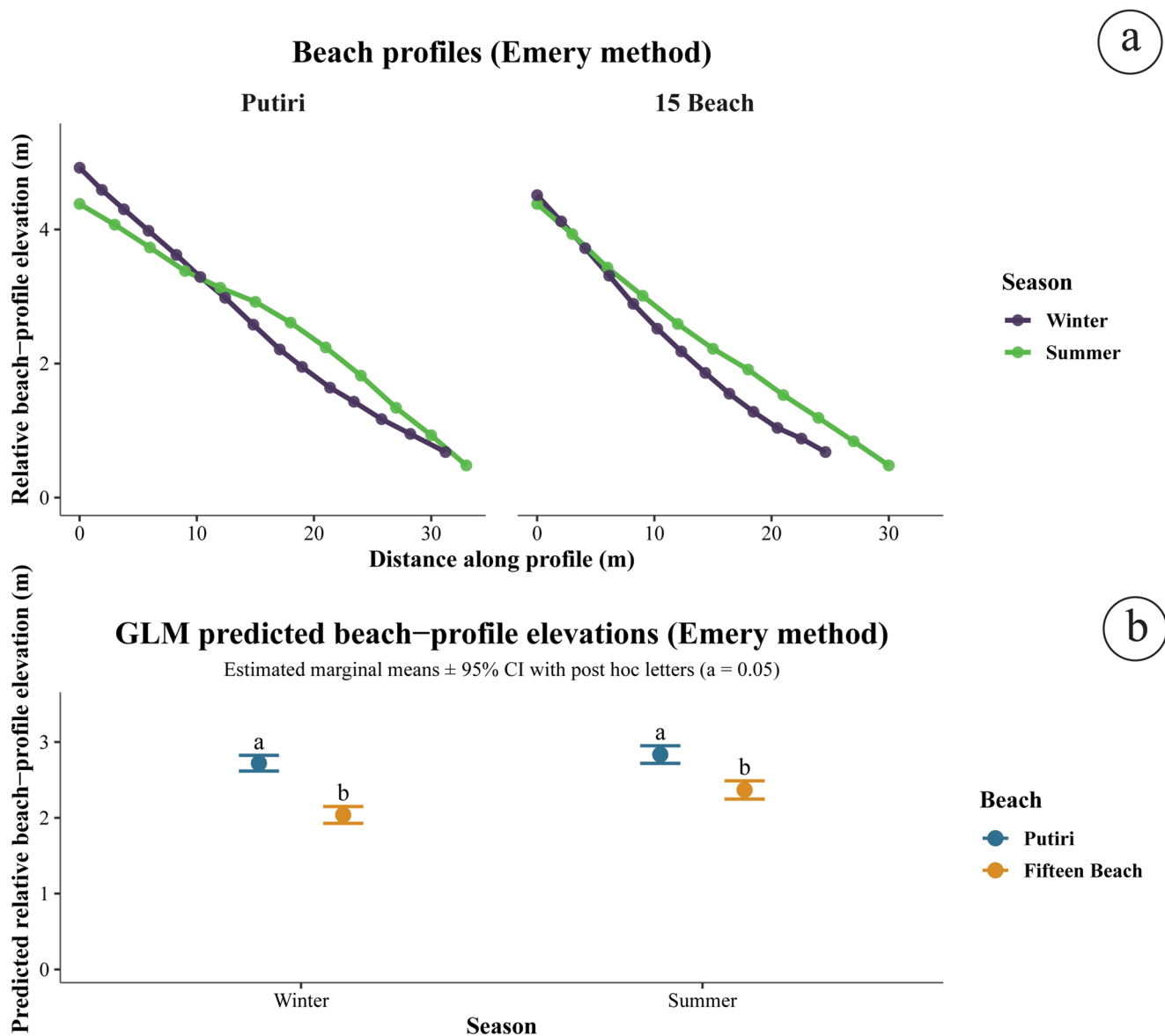


Fig. 2 Beach-scale morphodynamic characterization based on seasonal Emery beach-profile surveys. (a) Cross-shore beach profiles for Putiri Beach and 15 Beach in winter and summer, expressed as relative beach-profile elevation (m) along distance from the local profile reference point (m). Profiles were obtained using the Emery profiling method. (b) Generalized linear model (GLM) estimated marginal

means $\pm$ 95% confidence intervals of predicted relative beach-profile elevation by season for each beach. Different letters indicate significant post-hoc differences among beach $\times$ season combinations at $\alpha = 0.05$. The vertical scale has been corrected to represent relative elevation in meters

had significantly lower relative beach-profile elevation than the other beach-season combinations ($\alpha = 0.05$), whereas Putiri and 15 Beach did not differ detectably in winter at the mean profile distance. Sediment grain-size composition was dominated by sand-sized fractions across all beach-season combinations (Fig. 3a; Supplementary Table S5). Mean grain size remained within the fine-sand class for both beaches and both seasons. However, lower phi values in summer indicated relative coarsening within the fine-sand range rather than a categorical shift to medium or coarse

sand. Mean grain size was 2.79 ± 0.02 phi at Putiri in winter and 2.38 ± 0.41 phi in summer, while 15 Beach showed 2.85 ± 0.18 phi in winter and 2.37 ± 0.64 phi in summer. Fine sand was the dominant fraction in all combinations, representing 74.66% of the sediment mass at Putiri in winter, 54.02% at Putiri in summer, 78.49% at 15 Beach in winter, and 57.86% at 15 Beach in summer. Medium sand was the second most important fraction, ranging from 13.76% at 15 Beach in winter to 25.12% at Putiri in summer. Coarse and very coarse fractions increased in summer, supporting



Fig. 3 Sediment granulometry and wrack-algal composition across beaches and seasons. **(a)** Granulometric composition by sediment size class, from mud to gravel, for each beach $\times$ season combination. Cell colour and bubble size represent the total mass (g) of each sediment fraction. **(b)** Composition of stranded algal and vegetation wrack by functional group (green, brown, and red algae, terrestrial plants, and

other algae) for each beach $\times$ season combination. Cell colour and bubble size represent biomass (g m^{-2}). In both panels, letters inside cells indicate GLMM Tukey post-hoc groupings at $\alpha = 0.05$; groups sharing a letter are not significantly different. Together with Fig. 2, this figure provides the beach-scale sedimentological and ecological context supporting interpretation of local coastal morphology

a seasonal increase in the relative contribution of coarser particles within an overall fine-sand sediment matrix.

Granulometric descriptors further supported this interpretation (Supplementary Table S5). Winter sediments were moderately well sorted at both beaches, whereas summer

sediments were moderately sorted, indicating reduced sorting during summer. Skewness also varied among beach–season combinations, with Putiri winter classified as coarse-skewed, Putiri summer and 15 Beach summer as very coarse-skewed, and 15 Beach winter as very

fine-skewed. Because sediment size fractions are compositional, these patterns are interpreted primarily as changes in overall sediment texture, granulometric descriptors, and the relative contribution of major sediment fractions, rather than as fully independent changes in each grain-size class.

Wrack biomass and composition differed between beaches and seasons (Fig. 3b). Overall, stranded material was dominated by marine macroalgae (green, brown, and red algae) and terrestrial plant debris, but the relative importance of these components shifted by beach–season combination. At 15 Beach, total wrack biomass increased ~4-fold in summer ($219.52 \pm 83.31 \text{ g m}^{-2}$) compared with winter ($56.93 \pm 12.55 \text{ g m}^{-2}$). This summer peak was driven primarily by brown algae ($96.36 \pm 47.43 \text{ g m}^{-2}$) and substantial terrestrial plant debris ($76.93 \pm 64.84 \text{ g m}^{-2}$), with additional contributions from red algae ($32.41 \pm 22.30 \text{ g m}^{-2}$), green algae ($9.14 \pm 4.53 \text{ g m}^{-2}$) and other algae ($4.69 \pm 2.14 \text{ g m}^{-2}$). In winter at this beach, biomasses were lower across groups (brown: 21.52 ± 15.45 , red: 22.18 ± 12.02 , terrestrial: 12.22 ± 15.44 , green: 0.98 ± 0.57 , other: $0.04 \pm 0.05 \text{ g m}^{-2}$). At Putiri, total wrack biomass was comparatively stable across seasons (winter: $53.81 \pm 7.35 \text{ g m}^{-2}$; summer: $54.55 \pm 1.05 \text{ g m}^{-2}$), but composition shifted: summer deposits were dominated by terrestrial plants ($34.94 \pm 6.39 \text{ g m}^{-2}$) and brown algae ($12.49 \pm 1.92 \text{ g m}^{-2}$), whereas winter showed a more even distribution among brown algae ($18.55 \pm 4.04 \text{ g m}^{-2}$), terrestrial plants ($20.05 \pm 3.04 \text{ g m}^{-2}$) and red algae ($12.28 \pm 9.91 \text{ g m}^{-2}$), with consistently low green algae (winter: 2.63 ± 1.36 ; summer: $2.09 \pm 1.48 \text{ g m}^{-2}$) and other algae (winter: 0.31 ± 0.21 ; summer: $2.54 \pm 0.58 \text{ g m}^{-2}$). GLMM post-hoc groupings ($\alpha=0.05$) indicated that beach $\times$ season differences were group-dependent, consistent with a strong interaction structure in wrack deposition patterns.

A small number of taxa dominated macrofaunal abundance across beaches and seasons. Across all samples ($N=5450$ individuals), Talitridae accounted for 4067 individuals (74.6%), Oligochaeta for 1117 (20.5%), and Cirolanidae for 181 (3.3%) (together 98.4% of total abundance) while all remaining taxa each contributed $\leq 0.3\%$ and occurred patchily among stations (Fig. 4). Cirolanidae were recorded among the wrack-associated beach macrofauna sampled from stranded macrophyte deposits, but their origin as resident beach fauna or organisms transported with the wrack was not determined. The one-factor PERMANOVA detected significant differences in macrofaunal assemblage composition among beach–season combinations ($df=3$, pseudo- $F=4.42$, $p<0.05$). The absence of Hippoidea from the heatmap (Fig. 4) should not be interpreted as regional absence; mole crabs were not targeted by the sampling design because no specific lower-intertidal/swash-zone coring or excavation was performed. Pairwise tests

showed that all comparisons were significant after adjustment ($p_{\text{adj}} \leq 0.0144$) except 15 Beach summer vs. Putiri winter ($p_{\text{adj}}=0.087$) (Fig. 5). In terms of total density (individuals per station), abundance peaked at 15 Beach in winter ($196.2 \pm 207.9 \text{ ind station}^{-1}$; total=3531), compared with 15 Beach summer (52.2 ± 51.5 ; total=940), Putiri winter (41.9 ± 50.6 ; total=754), and Putiri summer (12.5 ± 17.3 ; total=225). At the aggregated level (major faunal groups), amphipods dominated most strongly at 15 Beach in winter ($80.63 \pm 16.58\%$ of individuals), consistent with the very high Talitridae counts ($176.9 \pm 216.1 \text{ ind station}^{-1}$; total=3184). In contrast, summer assemblages showed higher contributions of oligochaetes and insects, particularly at Putiri (summer: Oligochaeta $34.62 \pm 14.39\%$, Insecta $22.52 \pm 9.01\%$; winter: Oligochaeta $36.26 \pm 30.64\%$, Insecta $20.16 \pm 2.89\%$), while at 15 Beach summer oligochaetes and insects were also elevated (Oligochaeta $40.64 \pm 19.68\%$, Insecta $27.84 \pm 6.09\%$) relative to winter (Oligochaeta $12.15 \pm 12.13\%$, Insecta $4.22 \pm 3.54\%$) (Fig. 5).

Taken together, the beach-profile GLM, sediment and wrack GLMMs, granulometric descriptors, and Bray–Curtis PERMANOVA indicate that 15 Beach in summer represented the most distinctive beach–season condition. This combination was characterized by lower relative beach-profile elevation and a distinct cross-shore profile configuration, rather than by a quantitatively demonstrated reduction in beach slope. Sediments remained within the fine-sand class, but differed in sorting, skewness, and the relative contribution of major sediment fractions, as summarized in Supplementary Table S5. This beach–season combination also showed increased wrack biomass dominated by brown algae and terrestrial material, together with a distinct macrofaunal assemblage structure. In contrast, winter conditions at both beaches were associated with higher relative beach-profile elevations and strong fine-sand dominance, although differences in wrack composition and the relative dominance of key faunal groups remained evident. Together, these results show that beach–season combinations differed in relative beach-profile configuration, sediment texture, wrack subsidies, and macrofaunal assemblage composition. Although these analyses do not constitute a formal joint test of environmental covariation, the parallel patterns support the interpretation that seasonal physical conditions and organic subsidies are important contextual drivers of macrofaunal assemblage structure.

Discussion

This study integrates seasonal sand beach morphodynamics, sediment grain size, and wrack-algal composition to explain station-scale shifts in intertidal macrofaunal assemblages on

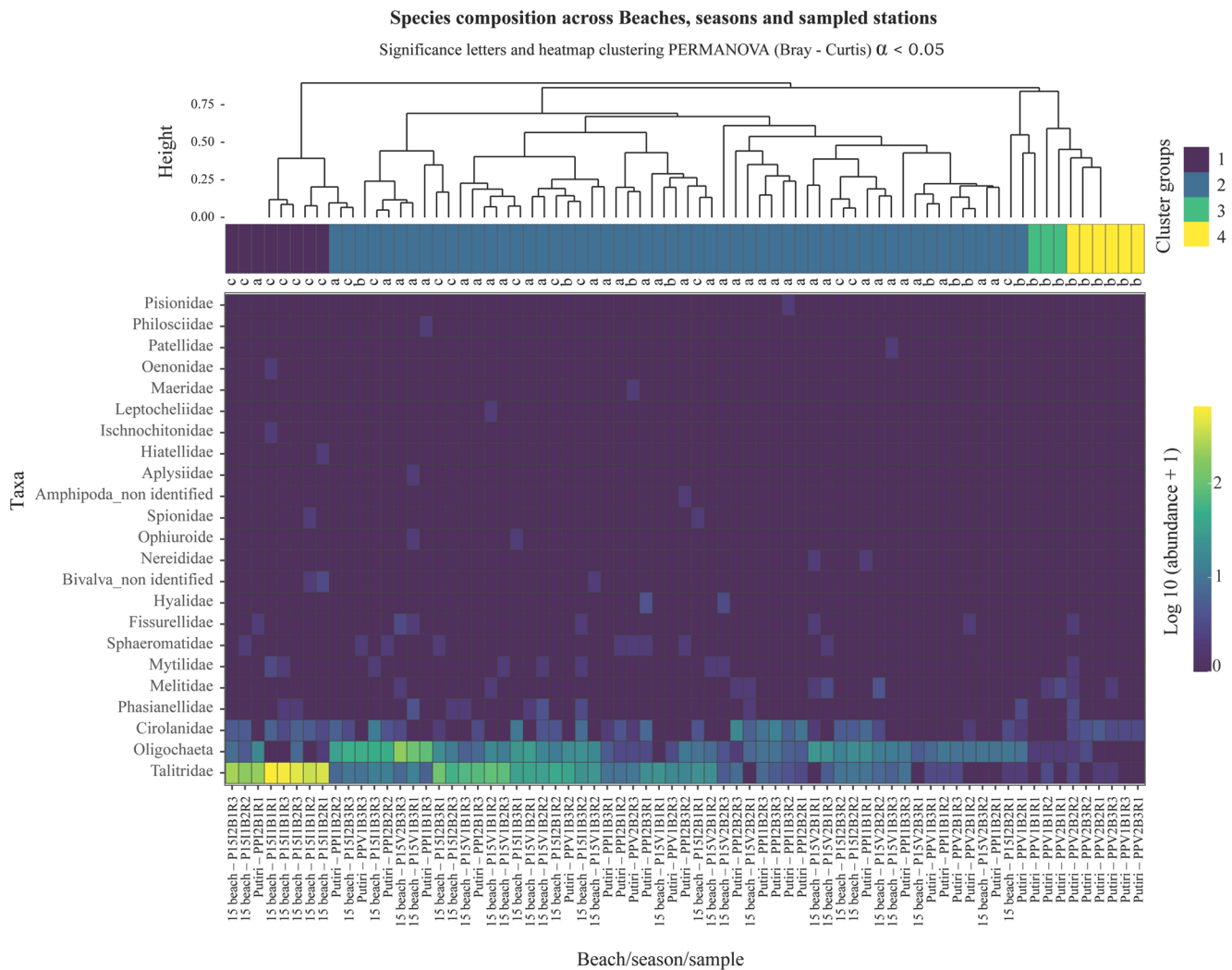


Fig. 4 Clustered heatmap of taxon abundance per station, based on $\log_{10}(\text{abundance} + 1)$ -transformed counts. Columns represent sampling stations nested within beach $\times$ season combinations, and rows represent macrofaunal taxa. Columns were hierarchically clustered using Bray–Curtis dissimilarities and average linkage clustering, with the upper dendrogram summarizing similarity among stations. The

coloured bar above the heatmap indicates station cluster membership. Letters associated with stations indicate the beach $\times$ season group to which each station belongs, based on pairwise PERMANOVA comparisons among beach $\times$ season combinations, with p-values adjusted using the Benjamini–Hochberg procedure. Groups sharing the same letter were not significantly different in assemblage composition

two tropical sandy beaches of the central Brazilian coast. Across beaches and seasons, the coupled physical–biological gradients observed here are consistent with the view of sandy shores as dynamic ecosystems where beach-profile dynamics, sediment grain size, and trophic subsidies jointly regulate community structure and ecosystem functioning (Wright and Short 1984; Defeo and McLachlan 2005; Hyndes et al. 2022). Below, we interpret the main patterns in relation to sandy-beach morphodynamic model, seasonal profile variability, sediment–fauna linkages, and the ecological role of wrack, while recognizing that both beaches fall within the intermediate morphodynamic range based on Dean’s parameter (Wright and Short 1984; Defeo and McLachlan 2005, 2013; Dugan et al. 2003; Hyndes et al.

2022). We then discuss global comparisons and management implications, particularly in light of increasing interest in harvesting beach-cast algae for industrial fertilizer production. An important clarification is that Putiri and 15 Beach were not classified as different morphodynamic states. Both beaches fell within the intermediate range according to Dean’s dimensionless fall-velocity parameter. Accordingly, the Emery profiles are interpreted here as descriptors of seasonal beach-profile morphology and relative elevation, not as independent evidence for contrasting morphodynamic states. Because beach profiles can vary seasonally or following short-term events such as storms, our interpretation focuses on beach–season associations among profile configuration, sediment texture, wrack subsidies, and macrofaunal

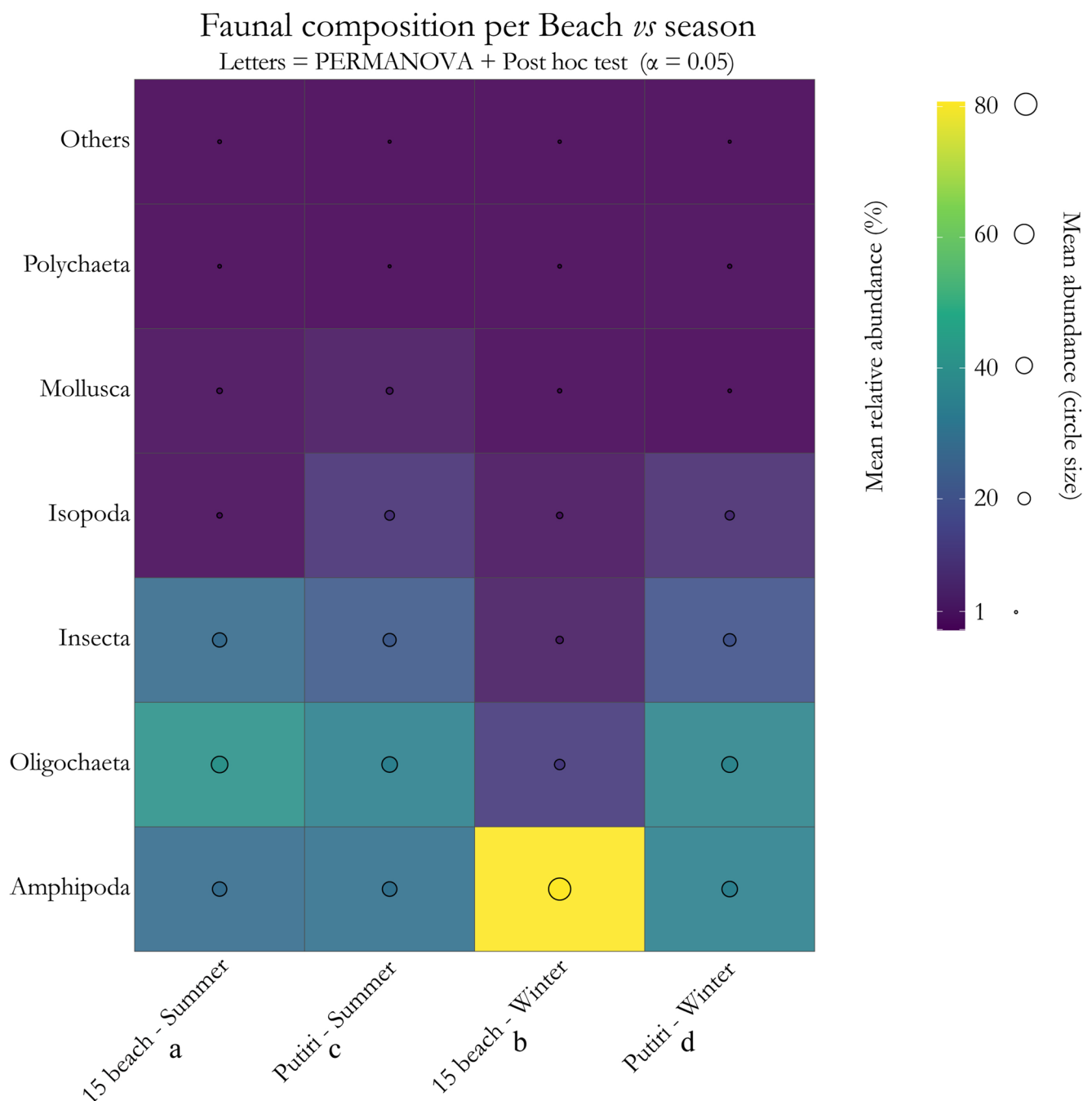


Fig. 5 Mean macrofaunal composition by beach and season. Heatmap showing the mean relative abundance (%) of major faunal groups for each beach $\times$ season combination: 15 Beach–summer, Putiri–summer, 15 Beach–winter, and Putiri–winter. Cell colour represents mean relative abundance, and circle size represents mean abundance, as indicated by the legends. Letters beneath each beach $\times$ season category summarize pairwise PERMANOVA groupings among beach $\times$ season combinations, with p-values adjusted using the Benjamini–Hochberg procedure. Categories sharing the same letter were not significantly different in overall macrofaunal assemblage composition

assemblage structure. Seasonal differences in the Emery profiles suggest that the two beaches differed in relative cross-shore profile configuration among sampling periods. However, because beach slope, tidal-stage-standardized elevation, nearshore hydrodynamics, and storm history were not directly quantified, these profile differences should be

interpreted cautiously as descriptive patterns rather than as direct evidence of contrasting morphodynamic processes. In sandy-beach systems, local wave climate, swash conditions, sediment mobility, beach orientation, and storm frequency can influence profile shape, wrack deposition, and sediment redistribution, producing seasonal variation

in beach morphology and habitat availability (Wright and Short 1984; Wright et al. 1979; Bernabeu et al. 2003; Defeo and McLachlan 2005). In the study area, previous oceanographic seasonal monitoring revealed important variation in wave height and temperature, with influence on coastal benthos of intertidal habitats (Mazzuco and Bernardino 2022; Mazzuco et al. 2026). In the present study, these processes provide plausible context for the observed beach-season patterns, but they should be regarded as potential mechanisms rather than demonstrated drivers. Accordingly, the observed beach-season differences are interpreted as associations among relative profile configuration, sediment texture, wrack subsidies, and macrofaunal assemblages.

The local coastal setting may also help explain some of the observed differences, particularly at 15 Beach. Rocky headlands and nearshore rocky outcrops occur close to this beach, as shown in the study-area map. Such features may influence local wave transformation, nearshore circulation, and the retention or accumulation of drifting macrophyte material. This could partly explain the higher wrack biomass observed at 15 Beach during summer, especially given that wrack deposition and persistence are known to depend on exposure, coastal orientation, hydrodynamic sorting, and local retention conditions (Kirkman and Kendrick 1997; Olabarria et al. 2007; Lastra et al. 2014; Hyndes et al. 2022). Nevertheless, because wave refraction, current patterns, and wrack transport pathways were not directly measured, the influence of rocky outcrops should be interpreted as a plausible hypothesis for future testing rather than as a demonstrated mechanism.

A similar caution applies to the interpretation of Emery beach-profile data. In this study, profile measurements were expressed as relative beach-profile elevation along each cross-shore transect, rather than as beach slope (%) or absolute elevation referenced to a fixed geodetic datum. Although this approach is useful for describing relative profile shape and beach-face configuration, profile elevation can be influenced by tidal stage, swash intensity, and the precise timing of measurements. Therefore, seasonal differences in relative beach-profile elevation should be interpreted as descriptive changes in local beach-profile configuration, rather than as direct quantitative evidence of absolute accretion, erosion, sediment-volume change, or morphodynamic-state shifts. Future studies should ideally combine repeated Emery profiles with standardized tidal-stage sampling, beach-slope estimates, and geodetic elevation surveys to better separate seasonal profile change from short-term hydrodynamic variability. The texture of sediment acts as a first-order filter for the macrofauna of sandy beaches because grain size and sorting modulate the volume and connectivity of interstitial spaces, sediment compaction, oxygen penetration, and the energetic cost of burrowing, thereby structuring patterns of

occupation and dominance among taxa (Defeo and McLachlan 2005, 2013). In this study, Folk and Ward granulometric descriptors indicated that both beaches were classified as fine sand in both seasons, although lower phi values in summer reflected relative coarsening within the fine-sand range, together with reduced sorting and predominantly coarse-skewed distributions. These results support seasonal variation in sediment texture and in the relative contribution of coarser fractions among beach–season combinations. However, we interpret these patterns cautiously and do not infer specific sediment-transport mechanisms from granulometry alone. Accordingly, we avoid inferring specific sediment-transport mechanisms from the granulometric data alone and restrict our interpretation to the observed dominance of fine sand, seasonal coarsening, and variation in the relative abundance of coarser fractions. In coarser sediments, greater porosity and permeability tend to favor water circulation and, therefore, greater oxygen availability for macrofauna; on the other hand, during low tide these sediments can drain and dry more quickly, increasing desiccation stress. While finer sediments are often associated with greater interstitial water retention and higher organic matter content, they exhibit lower permeability, which can reduce oxygen renewal and alter microhabitat quality. These internal physicochemical differences in the sediment shape habitat suitability for different groups (e.g., small burrowers and interstitial-dependent taxa in fine, well-sorted sands versus taxa more tolerant of substrate mobility and larger pores when coarse fractions increase), and potentially influence the rate of algal wrack decomposition by regulating oxygenation, moisture, and microbial activity in the sediment (Defeo and McLachlan 2005). In tropical environments, where temperature is less seasonally limiting than in temperate systems, these granulometry-mediated controls can become even more relevant to seasonal turnover, interacting with hydrodynamics to determine recruitment and persistence of dominant groups (Coutinho and Bernardino 2017; Costa et al. 2019).

Beach-cast macrophyte wrack represents a major cross-ecosystem subsidy that can elevate consumer biomass, increase local diversity and create biogeochemical ‘hotspots’ through microbial processing and nutrient remineralization in otherwise low-productivity sandy sediments (Dugan et al. 2003; Rodil et al. 2019; Hyndes et al. 2022). In the present study, wrack biomass and functional composition varied among beach $\times$ season combinations, indicating that both oceanographic delivery and retention (e.g., by beach slope, driftline position, and hydrodynamic sorting) are likely important. Functionally, different wrack types (e.g., brown vs red vs. green algae, and terrestrial plant material) can differ in tissue toughness, biochemical composition and decomposition rates, which can alter the

timing and magnitude of subsidy delivery to detritivores and higher trophic levels (Krumhansl and Scheibling 2012; Hyndes et al. 2022). Accordingly, the association between wrack patterns and macrofaunal composition is consistent with bottom-up effects whereby wrack availability enhances local food resources and shelter, increasing the abundance, richness, and biomass of wrack-associated detritivores and scavengers that can influence broader assemblage structure (Dugan et al. 2003; Hyndes et al. 2022).

Critically, the emerging industrial interest in collecting algal wrack for fertilizer or other bioproducts must be evaluated against these ecosystem functions. Reviews of drifting and beach-cast macroalgae harvesting highlight that removing wrack can reduce habitat complexity and trophic subsidies, potentially lowering secondary production and biodiversity on the beach and adjacent dunes (Kirkman and Kendrick 1997; Hyndes et al. 2022). Recent syntheses further emphasize that management should treat wrack as ecological infrastructure, not simply as ‘waste’, and that any utilization should be designed to avoid compromising shoreline ecosystem services and resilience (Harris and Defeo 2022; Martins et al. 2024a, b).

Despite strong biogeographic contrasts among regions, several broad regularities observed on beaches worldwide help contextualize patterns from the central Brazilian coast. First, the tight coupling among morphodynamics, sediment traits and macrofaunal metrics is a general feature of sandy shores, with beach state providing an integrative descriptor of swash climate and habitat harshness that often predicts richness, abundance and biomass patterns (Defeo and McLachlan 2005; Barboza and Defeo 2015). Second, wrack subsidies can be pivotal across latitudes, driving localized increases in diversity and abundance and strengthening land-sea trophic connectivity, from temperate systems with strong kelp detritus inputs to subtropical and tropical shores where wrack derives from diverse macroalgal and seagrass sources (Dugan et al. 2003; Krumhansl and Scheibling 2012; Hyndes et al. 2022).

What appears distinctive in the present study is the interaction between beach identity and season in both physical state and biological responses, suggesting that even neighbouring beaches can express different seasonal trajectories in habitat template and subsidy retention. This reinforces the need for site-specific baselines and cautions against extrapolating ‘average’ seasonal responses across a region without accounting for local exposure, sediment supply, and nearshore morphology (Defeo et al. 2009; Schlacher et al. 2007). Sandy beaches are globally threatened by coastal development, grooming and mechanical cleaning, trampling, pollution, and climate-driven shoreline change and storm impacts (Brown and McLachlan 2002; Defeo et al. 2009; Schlacher et al. 2007). Climate change is projected to

intensify erosion hazards and alter wave climates in many regions, with potentially severe consequences for the persistence of sandy shore habitat and associated biodiversity (Vousdoukas et al. 2020). In this context, the present results provide a mechanistic basis for monitoring programs that jointly track (i) beach profiles and sediment fractions, (ii) wrack biomass and composition, and (iii) indicator macrofaunal groups that respond to both habitat and subsidy dynamics (Defeo and McLachlan 2005; Hyndes et al. 2022).

Sediment texture is widely recognized as an important habitat filter for sandy-beach macrofauna because grain size and sorting can influence sediment stability, interstitial space, moisture retention, permeability, oxygen renewal, and burrowing conditions, thereby structuring patterns of occupation and dominance among taxa (Defeo and McLachlan 2005, 2013). In this study, however, these sediment-mediated processes were not measured directly. Therefore, they are best interpreted as plausible ecological pathways through which the observed granulometric differences may influence macrofaunal assemblages. Folk and Ward granulometric descriptors indicated that both beaches remained within the fine-sand class in both seasons, although lower phi values in summer reflected relative coarsening within the fine-sand range, together with changes in sorting, skewness, and the relative contribution of major sediment fractions. These descriptors are now summarized in Supplementary Table S5. Thus, rather than indicating a categorical shift in sediment type, the granulometric results suggest relatively subtle seasonal changes in sediment texture that may have contributed to differences in macrofaunal assemblage structure together with wrack availability and local beach-profile configuration. In tropical environments, where temperature is often less seasonally limiting than in temperate systems, such sediment-mediated controls may interact with hydrodynamic exposure and organic subsidies to influence recruitment and persistence of dominant groups, although this mechanism remains inferred rather than directly demonstrated here (Coutinho and Bernardino 2017; Costa et al. 2019).

From a management perspective, the results support a precautionary approach to beach-cast wrack removal, particularly where wrack deposits provide habitat and trophic subsidies for intertidal macrofauna (Dugan et al. 2003; Rodil et al. 2019; Hyndes et al. 2022). Because this study was observational and temporally limited, it does not quantify the direct effects of wrack removal or define extraction thresholds. However, the association between wrack availability and macrofaunal assemblage patterns suggests that removal for fertilizer or other uses should not be treated as an ecologically neutral activity. Beach-cast macroalgal biomass should also be interpreted primarily as a labile allochthonous organic-matter subsidy rather than as a stable

carbon stock, because stranded wrack can undergo rapid microbial and faunal decomposition, nutrient release, and CO₂ production unless burial, preservation, or export to long-term reservoirs is demonstrated (Lastra et al. 2018). Any proposed use of beach-cast wrack should therefore be preceded by site-specific environmental assessment and, where appropriate, licensing procedures that consider seasonal variation, local beach morphology, wrack composition, macrofaunal assemblages, and the presence of nearby source or retention features such as rocky outcrops. Monitoring programs should ideally include no-removal reference areas, seasonal restrictions where necessary, limits on removal intensity, avoidance of heavy machinery in the active beach zone, and adaptive biological indicators to detect changes in wrack-associated fauna and sediment conditions (Defeo et al. 2009; Harris and Defeo 2022; Martins et al. 2024a, b). These measures are consistent with the broader recognition that sandy beaches provide “packages” of ecosystem services (support for biodiversity, nutrient cycling, connectivity with fisheries and recreation) that depend on the integrity of the active littoral zone, from the dunes to the surf zone (Harris and Defeo 2022). Three limitations should be considered when interpreting these results. First, temporal coverage is limited to two seasonal snapshots; extending sampling to multiple years and including event-based surveys following storms would improve inference on how episodic forcing shapes beach-profile morphology, wrack delivery, and community recovery. In addition, directional wave data, wind speed and direction, beach orientation, storm frequency, swash dynamics, and wrack residence time were not explicitly quantified here. Incorporating these variables in future studies would allow stronger mechanistic attribution of the observed beach $\times$ season patterns. Second, station replication and taxonomic resolution (including the ‘non identified’ categories) constrain the ability to attribute patterns to species-specific life histories; future work should prioritize higher-resolution taxonomy (including molecular barcoding where necessary) and quantify size structure and functional traits of dominant taxa (Wright and Short 1984; Defeo and McLachlan 2005).

Conclusions

This study shows that wrack-associated macrofaunal assemblages on two tropical sandy beaches of the central Brazilian coast differed among beach–season combinations during the sampling period. Both beaches were classified within the intermediate morphodynamic range; therefore, the observed biological differences are best interpreted in relation to seasonal associations with relative beach-profile configuration, sediment texture, organic-matter content, and

wrack availability/composition, rather than as evidence of contrasting morphodynamic states.

Sediments remained within the fine-sand class across beaches and seasons, although granulometric descriptors indicated variation in sorting, skewness, and the relative contribution of major sediment fractions. Wrack biomass and composition also varied between beaches and seasons, with 15 Beach in summer showing the highest wrack accumulation, dominated mainly by brown algae and terrestrial plant material. Macrofaunal assemblages were strongly dominated by Talitridae, Oligochaeta, and Cirolanidae, but their relative contributions differed among beach–season combinations.

Overall, the results support an integrated interpretation in which physical habitat characteristics and trophic subsidies act as complementary gradients associated with macrofaunal assemblage variation. Because the study was observational and temporally limited, these relationships should be interpreted as empirical associations rather than demonstrated causal mechanisms. Future studies combining longer temporal coverage, direct hydrodynamic measurements, repeated standardized beach profiles, and experimental or manipulative approaches would help clarify the processes linking beach morphology, sediment texture, wrack subsidies, and macrofaunal community structure.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s41208-026-01177-2>.

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Author Contributions Antônio Batista Anderson: Conceptualization, Methodology, Formal analysis, Data curation, Visualization, Writing original draft. André Vassoler: Writing – review and editing. Gabriel Coppo: Writing – review and editing. Layza Roxanne Santana de Lima: Writing – review and editing. Angelo Fraga Bernardino: Conceptualization, Supervision, Funding acquisition, Project administration, Writing – review & editing.

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Data Availability Data and code used in this study are available from the corresponding author upon reasonable request.

Declarations

Clinical Trial Number not applicable.

Competing Interests The authors declare no competing interests.

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